

A Direct Gaussian-Interval Counterexample to Weighted Torsion Concavity

Literature-implied answer and exact pointwise strengthening

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Status. Question 18 of Cordero-Erausquin–Eskenazis [1] has a negative answer. Marín Sola–Salerno [2] already refuted its torsional-rigidity consequence for centrally symmetric Gaussian sets. We give a shorter, strictly stronger pointwise obstruction in dimension one. This is not a priority claim.

1 Source question and literature status

Let μ be an even log-concave measure on $\mathbb{R}^n$, let K, L be symmetric convex sets, and let $u(\lambda, \cdot)$ solve

$$\mathcal{L}_\mu u = -1 \quad \text{on } \lambda K + (1 - \lambda)L, \quad u = 0 \quad \text{on the boundary.}$$

Question 18 asks whether $(\lambda, x) \mapsto \sqrt{u(\lambda, x)}$ is concave.

boundary-value problem (115) on $\lambda K + (1 - \lambda)L$, then the function $\sqrt{u(\lambda, x)}$ is concave in the pair of variables (λ, x) . Afterwards, inequality (114) becomes an immediate consequence of (116) and the concavity principle for the marginal (5) with $\beta = 2$. We refer, for instance, to [61, Section 4] for a comprehensive study of concavity properties of solutions to more general boundary-value problems.

In view of the recent interest in Brunn–Minkowski inequalities for measures of symmetric convex sets, it is worthwhile to investigate such inequalities for weighted analogues of other analytic functionals in the presence of symmetry. Given an even probability measure μ on $\mathbb{R}^n$, the μ -weighted torsional rigidity $\tau_\mu(K)$ of a symmetric convex set K is defined by (113), where all dx integrals are replaced by $d\mu(x)$. It can also be rephrased in terms of the solution to the boundary value problem

$$\begin{cases} \mathcal{L}_\mu u = -1, & \text{in } \text{int}(K) \\ u \equiv 0, & \text{on } \partial K \end{cases}. \quad (117)$$

We refer to [48, 32] for some recent investigations on the functional τ_μ . In view of the weighted concavity principle of Theorem 1, an affirmative answer to the following question would prove the weighted analogue of (114) for rotationally invariant log-concave weights.

Question 18. Fix an even log-concave measure μ on $\mathbb{R}^n$ and let K, L be symmetric convex sets in $\mathbb{R}^n$. Let also $u(\lambda, \cdot) : \lambda K + (1 - \lambda)L \rightarrow \mathbb{R}$ be the unique solution to (117) on $\lambda K + (1 - \lambda)L$, where $\lambda \in [0, 1]$. Is the function $\sqrt{u(\lambda, x)}$ a concave function of the pair of variables (λ, x) ?

Figure 1: Question 18 in the source PDF, page 26.

Faithful transcription. For an even log-concave μ and symmetric convex K, L , is $\sqrt{u(\lambda, x)}$ jointly concave when $u(\lambda, \cdot)$ is the unique weighted torsion solution on $\lambda K + (1 - \lambda)L$?

The source explains immediately before the question that an affirmative answer would imply the weighted Brunn–Minkowski inequality for torsional rigidity. In March 2026, Marín Sola and Salerno explicitly addressed [1, Question 18] and disproved that consequence for centered Gaussian balls in dimensions $n \geq 3$.

(Q2) Does the case of centrally symmetric sets allow us to say anything more?

In this paper, we answer these questions as follows. Referring to (Q1), we show with two examples that both concavity and convexity properties cannot be true in general for open, bounded convex sets. Concerning question (Q2), and addressing [14, Question 18], we provide a counterexample showing that the functional $T_\gamma^{1/(n+2)}$ is neither concave nor convex for centrally symmetric convex sets. Specifically, considering $\Omega_i = B_{R_i}$ (an Euclidean ball of radius R_i centered at the origin), $i = 0, 1$, we find that, in dimension $n \geq 3$, no concavity or convexity property holds for the Gaussian torsion with exponent $1/(n+2)$. This will be the content of Example 3.3.

In light of the latter, one may wonder about the smallest exponent α such that the functional T_γ^α is convex (or concave) for centrally symmetric convex sets. We provide a partial answer; we show that, for Euclidean balls, the functional $T_\gamma^{1/3}$ is convex with respect to the Minkowski addition. This is the content of the following theorem.

Figure 2: The later negative answer, arXiv:2603.19164, page 3.

2 Proof intuition

On a centered Gaussian interval the differential equation can be integrated once. The outward slope of the torsion function is a positive function $B(R)$ satisfying the exceptionally simple ODE $B' = 1 + RB$. Its derivative is itself strictly convex. This forces the square root of the torsion value at the center to be *strictly convex* as the radius grows, the opposite of the proposed concavity.

3 Exact pointwise counterexample

Let γ be the standard Gaussian measure on $\mathbb{R}$, so

$$\mathcal{L}_\gamma v = v'' - xv'.$$

For $R > 0$, let u_R be the unique solution

$$u_R'' - xu_R' = -1 \quad \text{on } (-R, R), \quad u_R(-R) = u_R(R) = 0.$$

Theorem 1. *The function*

$$G(R) = \sqrt{u_R(0)}$$

is strictly convex on $(0, \infty)$. Consequently, Question 18 is false already for $\mu = \gamma$, $K = [-1, 1]$, and $L = [-3, 3]$.

Proof. Evenness and uniqueness give $u_R'(0) = 0$. Multiplying the equation by $e^{-x^2/2}$ and integrating from 0 to x yields

$$u_R'(x) = -B(x), \quad B(x) := e^{x^2/2} \int_0^x e^{-s^2/2} ds.$$

The boundary condition at R therefore gives

$$F(R) := u_R(0) = \int_0^R B(t) dt.$$

Direct differentiation shows

$$B'(R) = 1 + RB(R), \quad B''(R) = R + (1 + R^2)B(R) > 0 \quad (R > 0).$$

Since $F' = B$ and $F'' = B'$, the sign of G'' is the sign of

$$H(R) := 2F(R)B'(R) - B(R)^2.$$

But $H(0) = 0$ and the cancellation $F' = B$ gives

$$H'(R) = 2F(R)B''(R) > 0 \quad (R > 0).$$

Thus $H(R) > 0$ and $G''(R) > 0$ for every $R > 0$.

Now $\frac{1}{2}K + \frac{1}{2}L = [-2, 2]$. Joint concavity in Question 18 would imply, at $x = 0$,

$$G(2) \geq \frac{G(1) + G(3)}{2},$$

whereas strict convexity gives the reverse strict inequality. □

4 Verification, scope, and review

The high-precision checker `code/verify_gaussian_intervals.py` independently integrates the displayed formula. Its output is

```
sqrt(u_1(0)) = 0.77184798925226050830374705903
sqrt(u_2(0)) = 2.12169800306949307422074845125
sqrt(u_3(0)) = 6.52492402508500893933530599964
midpoint concavity deficit = 1.52668800409914164959877807808
strict midpoint violation verified
```

Scope and limitations. This settles only Question 18. It does not resolve hereditary convexity (Question 15) or the negative-exponent weighted-marginal problem (Question 19). Eight focused upgrade routes were tried for Question 19: separable negative moments, a quadratic perspective, max-type functions, radial two-piece and multipiece polyhedra in dimensions up to twelve, anisotropic quadratic perspectives, smooth log-sum-exp couplings, and anisotropic log-concave weights. No negative Hessian survived certified quadrature.

Novelty boundary. The question is already answered negatively by [2]; the value here is the exact one-dimensional pointwise strengthening and elementary proof. A bounded search of the run indexes and primary arXiv sources found no earlier statement of this particular strict-convexity identity. No priority is claimed.

Human-review recommendation. High confidence. Check only the convention $\mathcal{L}_\gamma = \frac{d^2}{dx^2} - x \frac{d}{dx}$ and the logical specialization to centered intervals; all remaining steps are two differentiations and a strict sign.

References

- [1] D. Cordero-Erausquin and A. Eskenazis, *Concavity principles for weighted marginals*, arXiv:2506.16941 (2025).
- [2] F. Marín Sola and F. Salerno, *Remarks on Brunn–Minkowski-type inequalities related to the Ornstein–Uhlenbeck operator*, arXiv:2603.19164 (2026).